

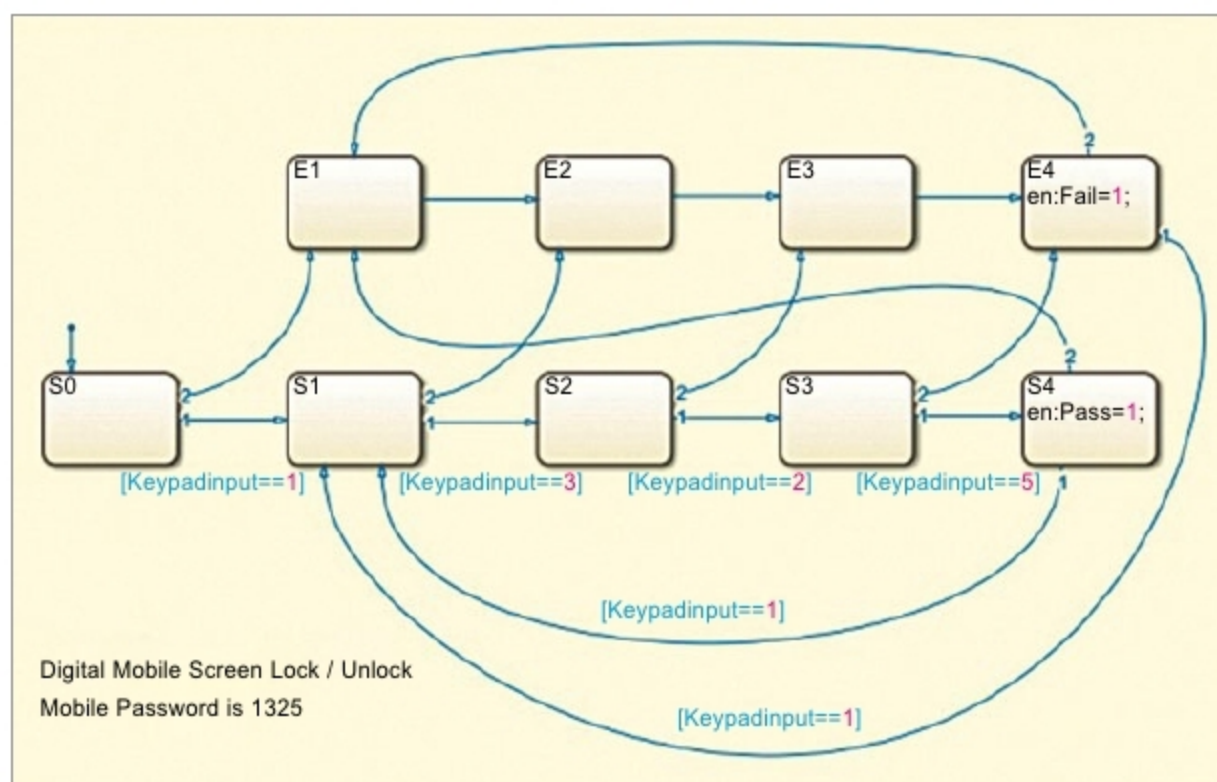


Fig. 3: Internal structure (chart) of simulation

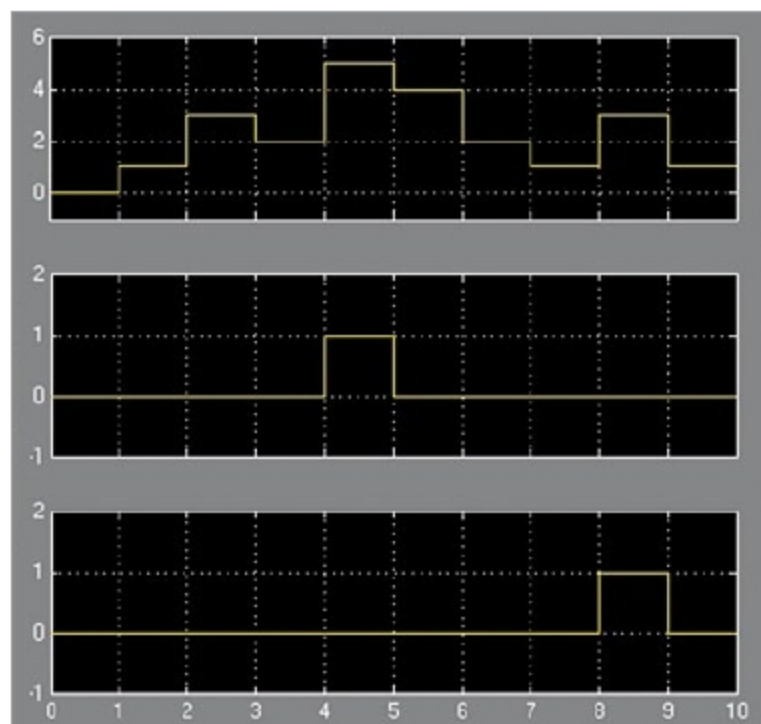


Fig. 4: Output waveform of simulation

9. In Model Explorer window, click on Add menu and select Data to add input (KeypadInput) and two output variables (Pass and Fail) to the chart.

10. To create a test stimulus in Signal Builder block for verifying the accuracy of the model, double-click on the block to open it. A new window will open, as shown in Fig. 2. Select Signal→Replace with→Custom... option, and enter y value as [0 0 1 1 3 3 2 2 5 5 4 4 2 2 1 1 3 3 1] and time value as [0 1 1 2 2 3 3 4 4 5 5 6 6 7 7 8 8 9 9].

11. Connect the three blocks as shown in Fig. 1 and run the simulation to view the results.

Testing and simulation results

1. Open MATLAB and go to Simulink Library Browser under File→Open and locate the .slx file (MOBILE-LockUnlock.slx).

2. After clicking on .slx file, a simulation window will open, which is the block diagram of the simulation (Fig. 1). It comprises three blocks, namely, input (Signal Builder), system (Chart) and output (Scope).

3. Double-click on Signal Builder block to see input. A waveform will open (Fig. 2).

This waveform indicates the stored password, of which one is correct (1 3 2 5) and the other is incorrect (4 2 1 3).

4. Click on Chart block to open the internal structure/programming. Here, you will see shells S and E, and the stored password (that is, 1 3 2 5), as shown in Fig. 3.

After designing the simulation, testing is required to ensure the cor-

rectness of the design. For testing the simulation, compile the block diagram. When compiling is done, open the output or scope by double-clicking it. Input test stimulus and outputs (Pass and Fail) will be displayed in the waveform window, as shown in Fig. 4.

Top waveform is the input test stimulus containing a series of values (0 1 3 2 5 4 2 1 3 1) in the form of waveforms that are fed as inputs to the application.

Middle waveform shows the correct password (1 3 2 5), which produces output value 1 at the fourth clock input, indicating successful unlocking of the Android phone screen.

Bottom waveform shows the wrong password (4 2 1 3), which produces output value 1 at the eighth clock input, indicating an unsuccessful attempt.

Steps for changing the password

From Signal Builder block, select Signal→Replace with→Custom... and a Custom Waveform dialogue box will appear. Here, you can change time values and y values as per the new password. Time values (also known as t values) will be on x axis and y values on y axis.

1 3 2 5 is the default password and 4 2 1 3 is a random incorrect password. Values of x-y waveform for 1 3 2 5 numbers are as given below.

T- values = [0 1 1 2 2 3 3 4 4 5 5 6 6 7 7 8 8]

Y- values = [0 0 1 1 3 3 2 2 5 5 4 4 2 2 1 1 3]

Note that, t values may remain constant. You just have to change y values according to your new password.

After putting waveform values, go to the chart and update the same password in keypad input (Fig. 3). Then go back to the block diagram of the simulation. Run the simulation and check output on the scope. Make sure you have entered the correct new password, or else you will not be able to get the pass output graph. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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